Code No: 136CF **R16**

JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD
B. Tech III Year II Semester Examinations, March - 2024
INTRODUCTION TO ANALYTICS
(Computer Science and Engineering)

Time: 3 Hours **Max. Marks: 75**

Note: i) Question paper consists of Part A, Part B.
ii) Part A is compulsory, which carries 25 marks. In Part A, Answer all questions.
iii) In Part B, Answer any one question from each unit. Each question carries 10 marks and may have a, b as sub questions.

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PART – A

(25 Marks)

1.a)	What is structured array? Give example.	[2]
b)	Define work management and its prioritization.	[3]
c)	Differentiate between Team work and Individual work.	[2]
d)	What are the types of variables available in R?	[3]
e)	What are the uses of NOSQL databases?	[2]
f)	How to Read R output in Excel? Explain.	[3]
g)	Define Multicollinearity and explain.	[2]
h)	With example, explain about forecasting.	[3]
i)	What are requirements gathering? Explain.	[2]
j)	Write the purpose of engineering design.	[3]

PART – B

(50 Marks)

2.a)	Discuss about various Data types in R with example.	
b)	With examples, explain about outliers and its significance.	[5+5]
	OR	
3.a)	Write and explain R functions and R loops with examples.	
b)	Explain about the concept of Reading Datasets.	[6+4]
4.a)	Explain about Teamwork communication and time management.	
b)	Discuss about probability distribution and its computation in R.	[5+5]
	OR	
5.a)	Explain about bivariate random variables in detail.	
b)	With an example in R, explain about Central Limit Theorem.	[5+5]
6.a)	Develop a R program to illustrate database connectivity.	
b)	Explain the features of NOSQL.	[5+5]
	OR	
7.a)	Explain about Excel and R integration with R connector.	
b)	Discuss about the term NOSQL. And differentiate between NOSQL and SQL.	[5+5]

8.a) Explain the concept of basic Regression Analysis with a suitable example.
b) Discuss about ANOVA test with example. [5+5]

OR

9.a) With suitable example, discuss about multiple regression and its applications.
b) Explain the terms correlation and auto correlation. Give examples. [5+5]

10.a) Discuss about the optimality function for a closed loop system.
b) Compare and contrast the manufacturing and production activities with respect to automotive technology. [5+5]

OR

11.a) Make a comparison of business analytics and organization decision-making processes.
b) What is business objective? How do you gather all the data related to business objective? Discuss. [5+5]

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